Ada Semantic Interface Specification (ASIS)
Frequently Asked Questions

What is ASIS?

The Ada Semantic Interface Specification (ASIS) is an interface between an Ada environment as defined by ISO/IEC 8652 (the Ada 95 Reference Manual) and any tool requiring information from this environment. An Ada environment includes valuable semantic and syntactic information. ASIS is an open and published callable interface which gives CASE tool and application developers access to this information. ASIS has been designed to be independent of underlying Ada environment implementations, thus supporting portability of software engineering tools while relieving tool developers from having to understand the complexities of an Ada environment's proprietary internal representation.

What type of Tools can be developed with ASIS?

Examples of tools that benefit from the ASIS interface include: code restructuring tools, code browsing and navigation tools, coding style and standards compliance tools, cross-reference tools, data flow analysis tools, dependency tree analysis tools, design tools, document generation tools, invocation (call) tree analysis tools, language-sensitive editing and pretty-printing tools, language translation tools, quality assessment tools, reverse engineering tools, re-engineering tools, safety and security compliance tools, static correctness verifiers, tasking analysis tools, test-case generation and coverage analysis tools, and usage, quality, and complexity metrics tools. In fact most of these have already been developed using the ASIS interfaces by people in Australia, Canada, China, France, Germany, Japan, Russia, Sweden, Switzerland, the United Kingdom, and the United States.

How would a CASE Tool use ASIS?

For example, ASIS can be used by a metrics tool to evaluate Ada application code. The symbolic name, type, and usage of each object declared can be obtained through the ASIS interface to support the requirements of the metrics tool. Use of the object can be identified by traversal of a logical semantic tree using the ASIS interfaces. Counts for the numbers of each desired occurrence can then be presented to users of the metrics tool. Examples of commercial tools which use ASIS include: Ada Analyzer by Little Tree Consulting, Adamat by DRC, AdaQuest by GRC, AdaWise by Odyssey Research Associates, and ReEngineer by Loral Defense Systems.

How might an Application use ASIS?

The *Universal Delogger* was built using ASIS for an Air Force satellite system called "System 1" to analyze (delog) messages sent from the satellite to a ground station. The satellite communicates to the ground station via messages in a byte-stream. An earlier delogger implementation required a rigorous interface protocol and specialized processing for each possible message. Instead of a rigorous protocol, the message header provides a number which maps to an Ada record type. The remaining portion of the message byte-stream contains the composite value of that record. When the
message is received, the *Universal Delogger* assigns portions of the byte stream to named components using ASIS and the optional ASIS data decomposition interfaces. The named components with their values are then presented to an operator. Developmental cost benefit: Many, but most notable is that each major build had to rebuild the message dictionary; the time required to build this "types dictionary" was reduced from 704 hours to 2 hours for each rebuild.

Why Use ASIS to get Information from the Ada Environment?

Tools needing syntactic and semantic information would otherwise have to recompute the information from all the source code (which could take a long time and require the semantic analysis of a compiler), fake the information (dangerous when reliable information is needed), or access the information through proprietary interfaces to an implementor's Ada environment (non-portable and breaks when proprietary interfaces change as they frequently do from one release to the next). ASIS provides a highly portable, easy-to-use interface, insulating users from changes in proprietary representations.

What is the Current ASIS Specification?

The current ASIS specification is Version 1.1.1 to Ada 83 (ISO Ada 87). This version has been stable since June 1994. It comes complete with a test suite so vendors can develop ASIS interfaces that meet the ASIS specification. Tool vendors can also use the test suite to assess the risk of porting a tool from one Ada vendor's environment to another Ada vendor's environment. ASIS is evolving for Ada 95. A draft Committee Draft is available on the ASIS Home Page.

How can I learn more about ASIS?

An ASIS home page on the World Wide Web contains all the latest information on ASIS. It also provides access to highlights of recent ASISWG/ASISRG meetings and several publicly available ASIS Tutorials.

ASIS is on World Wide Web => *http://www.acm.org/sigada/wg/asiswg*

How can I get the Current ASIS Specification?

The ASIS World Wide Web home page makes available all publicly available ASIS information, including the current ASIS Specification. The ASIS specification is also available via anonymous ftp from the *sw-eng.falls-church.va.us* host in the *public/AdaIC/work-grp/asiswg/asis/v1.1.1* directory. One should login as anonymous with your email address as your password. The Ada Information Clearinghouse (AdaIC) [+1 (703) 681-2466] can assist those new to using ftp. Those not on Internet can access ASIS via the AdaIC Bulletin Board at +1 (703) 681-2845 using any auto Baud rate to 28,800, Data Bits = 8; Parity = None; Stop Bits = 1, Xon/Xoff flowcontrol. One should login as guest with the password Ada4sw-eng.

What About ASIS for Ada 95?

An ASIS for Ada 95 is in process. This version is called ASIS 95. It represents an evolution of ASIS 83 to support all the enhancements within Ada 95. ASIS 83 provided an interface to the Ada 83 Program Library. The Ada Program Library is drastically different for Ada 95. These libraries can be extensive as with many implementations in Ada 83 or non-existent. ASIS 95 is better characterized as an interface to the Ada environment. Element kinds and ASIS interfaces have been developed for ASIS 95. A rather complete draft for the ASIS 95 specification was distributed at Tri-Ada'95 (ASIS 2.0.E); a draft Committee Draft has been submitted WG9 this summer based on (ASIS 2.0.G). An ASIS 95 implementation to the Free Software Foundation Ada 95 GNAT compiler is being developed by Dr. Sergey Rybin from Moscow State University in conjunction with Professor Alfred Strohmeier of the Swiss Federal Institute of Technology. A prototype ASIS 95 implementation is available by anonymous ftp on the *lglftp.epfl.ch* host in the *pub/ASIS* directory. It is a prototype and has components that are far from being completed. Several Ada 95 compiler vendors will have an ASIS 95 implementation in the near future.

Will ASIS work with Every Ada Vendor?

ASIS is designed to be implemented on a variety of machines and operating systems by a large number of Ada Vendors and to support Ada semantic requirements for a wide range of client tools. The services provided in ASIS are mainly primitive services with the intent that many layers of secondary higher level sophisticated services addressing the variant needs of specific Integrated Software Engineering Environment (ISEE) tools can be created from these primitives. If ASIS is not available for your favorite hardware and operating system, you might talk to your vendor about providing ASIS interfaces in the near future.

Where did ASIS originate?

ASIS was initiated as a STARS activity in 1989. It became unfunded for FY90 when the STARS program decided not to fund standardization activities. Several vendors continued work in ASIS on their own. In 1992 the Ada Board recognized the potential benefits to the Ada community of an ASIS standard and recommended that the Ada Joint Program Office (AJPO) director support by whatever means possible the development of an ASIS standard and its submission to ISO/WG9 for publication. Currently all funding comes from the Ada community. The Association for Computing Machinery (ACM's) Special Interest

Group on Ada (SIGAda) is now continuing this important work though volunteer effort in the ASIS Working Group (ASISWG). An ASIS Rapporteur Group (ASISRG) was established on 28 April 1995 by ISO/IEC JTC1/SC22 WG9 to standardize ASIS as an international standard for Ada 95 [ISO/IEC 8652.1995
(15 February 1995)]. On 28 May 1996, JTC1 approved the project, assigning NP 22-15291 to the standardization effort.

How do I get involved with ASISWG/ASISRG?

The ASISWG and ASISRG intend to evolve ASIS for the Ada 83 (ISO Ada 87) interfaces into ASIS for Ada 95. The result will be an international ISO/IEC standard complementing the Ada 95 standard. Your comments and participation are welcome. The following electronic mail forums now exist for the ASISWG/ASISRG:

1. For high-level non-technical discussions => *asis@sw-eng.falls-church.va.us*
   To subscribe, send email to: *asis-request@sw-eng.falls-church.va.us*

2. For detailed technical discussions => *asis-technical@sw-eng.falls-church.va.us*
   To subscribe, send email to: *asis-technical-request@sw-eng.falls-church.va.us*

Subscribe with your preferred email address, name, telephone number, and surface mail address. For other questions, contact Mr. Currie Colket, Chair of ASISWG/ASISRG, *colket@smtp-gw.spawar.navy.mil*, +1 (703) 602-1483, or Mr. Clyde Roby, Recorder of ASISWG, *roby@ida.org*, +1 (703) 845-6666.

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